

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

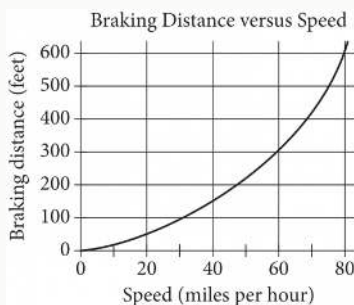
Problem-Solving and Data Analysis

12

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS



The graph above shows the relationship between the speed of a particular car, in miles per hour, and its corresponding braking distance, in feet. Approximately how many feet greater will the car's braking distance be when the car is traveling at 50 miles per hour than when the car is traveling at 30 miles per hour?

- A. 75
- B. 125
- C. 175
- D. 250

Shaquan has 7 red cards and 28 blue cards. What is the ratio of red cards to blue cards that Shaquan has?

- A. 1 to 4
- B. 4 to 1
- C. 1 to 7
- D. 7 to 1

A table of the US minimum wage for 6 different years is shown below.

Year	US minimum wage (dollars per hour)
1960	1.00
1970	1.60
1980	3.10
1990	3.80
2000	5.15
2010	7.25

What was the percent increase of the minimum wage from 1960 to 1970?

- A. 30%
- B. 60%
- C. 62.5%
- D. 120%

Number of High School Students Who
Completed Summer Internships

High school	Year				
	2008	2009	2010	2011	2012
Foothill	87	80	75	76	70
Valley	44	54	65	76	82
Total	131	134	140	152	152

The table above shows the number of students from two different high schools who completed summer internships in each of five years. No student attended both schools. Of the students who completed a summer internship in 2010, which of the following represents the fraction of students who were from Valley High School?

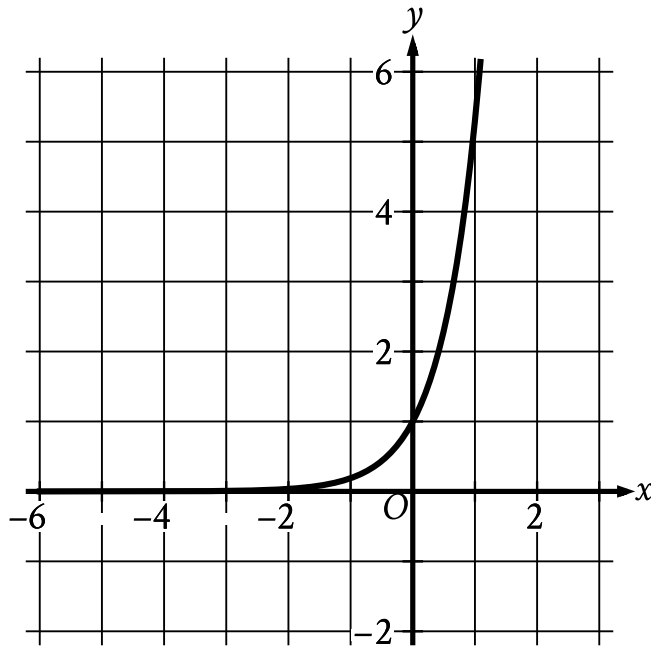
- A. $\frac{10}{140}$
- B. $\frac{65}{140}$
- C. $\frac{75}{140}$
- D. $\frac{65}{75}$

71, 72, 73, 76, 77, 79, 83, 87, 93

What is the median of the data shown?

- A. 71
- B. 77
- C. 78
- D. 79

The graph of the function f is shown, where $y = fx$.



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point (0 comma 1).
 - In quadrant 1, the curve trends up sharply.
- As x decreases, the curve approaches the line y equals 0.
- The curve passes through the following points:
 - approximately (negative 1 comma 0.2)
 - (0 comma 1)
 - approximately (1 comma 5.3)

Which of the following best describes the function f ?

A. Decreasing exponential

- B.** Increasing exponential
- C.** Decreasing linear
- D.** Increasing linear

A group of monarch butterflies migrated from Chicago, Illinois, to Michoacán, Mexico, flying a total of 2,100 miles. It took a single butterfly in the group 120 days to travel this route one way. On average, how many miles did the butterfly travel per day?

- A. 0.057
- B. 0.729
- C. 17.5
- D. 24

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

There are 250 trees in a park. Of these trees, 6% are birch trees. How many birch trees are in the park?

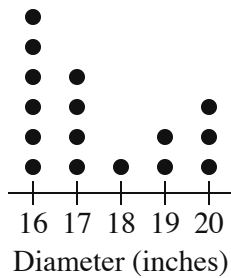
- A. 6
- B. 15
- C. 75
- D. 244

	Live east of the river	Live west of the river	Total
Less than 40 years old	17	11	28
At least 40 years old	18	89	107
Total	35	100	135

The table summarizes members of a local organization by age and whether they live east or west of the river. If a member of the organization is selected at random, what is the probability that the selected member is at least 40 years old?

- A. $\frac{28}{135}$
- B. $\frac{35}{135}$
- C. $\frac{100}{135}$
- D. $\frac{107}{135}$

To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



- The data for the dot plot are as follows:
 - 16: 6 dots
 - 17: 4 dots
 - 18: 1 dot
 - 19: 2 dots
 - 20: 3 dots

Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of 16 inches?

- A. 16
- B. 6
- C. 4
- D. 1